

June-July Exam session Projects List

Objectives

The goal of the project is to tackle one of the proposed topics in the field of Computer Vision, developing non-trivial solutions. Students should explore the problem from an original perspective, applying methods that go beyond conventional solutions and demonstrating critical thinking and problem-solving skills.

Deadline

Students willing to deliver their project in the June-July exam session must fill [this Google Form](#) by May 10th.

Lines of conduct

- **Student groups:** The project can be carried out by a group, each consisting of a maximum of 3 people. Projects can also be completed by individual students, but we suggest to work in team.
- **Notebook format:** The project must be implemented using a notebook (e.g., Google Colab, Kaggle) or with an IDE (e.g., VSCode, PyCharm). The code should be optimized to support GPU usage and run without any error. The delivered code must follow the structure outlined below:
 - *Imports:* all the needed packages (for the notebook format)
 - *Globals:* useful variables on the whole code
 - *Utils:* code support functions
 - *Data:* everything related to data management
 - *Network:* code to structure the neural network
 - *Train:* part containing the training cycle elements
 - *Evaluation:* tests needed for the trained network

You can find a sample template at this [link](#). Try to maintain as much as possible this conceptual structure.

- **Deep learning framework:** All projects **MUST** be done in Python via the Pytorch framework.
- **Project assignment:** You are required to choose a project through [this Google Form](#). In this form, you will provide information about your team and the project, and include the link to the project's GitHub repository. In this repository you have to upload:
 - Code (or notebook) implementing the project
 - Dataset (or a link to it)
 - Project presentation
 - Detailed README to provide a quick overview of the project and instructions on how to run it
- **Project submission:** The project must be presented on one of the exam dates. It can be presented at a different time than the written exam. Both the written exam and the project **MUST** be completed within the academic year (i.e., between the June 2026 session and the March 2027 session).
- **Plagiarism:** Any attempt to plagiarize, whether by copying other students' work, directly replicating code from online resources, or submitting content highly retrieved from generative AI models, will be strictly penalized. This course values originality and personal effort; therefore, students must submit independently developed solutions. On the other hand, it is acceptable to consult external resources for inspiration or guidance. Violations will be subject to academic disciplinary actions, potentially including a failing grade for the project or further measures outlined by university policy.

Project-X: Student Project Proposal through Advanced Computer Vision

Abstract: Computer Vision is a vast and rapidly evolving field, with novel architectures and paradigms being introduced continuously. This open-ended project empowers students to tackle an advanced, self-selected topic that goes beyond standard curriculum boundaries. Rather than solving a pre-defined problem, students are required to act as researchers: they must identify a specific gap, flaw, or limitation in current state-of-the-art literature and propose an original, ad-hoc methodological improvement to address it. This project demands critical thinking, deep literature analysis, and the ability to design and validate an original algorithmic or architectural novelty in a complex Computer Vision domain.

Dataset: Student-selected (Must be strictly open-source and publicly available; e.g., via Kaggle, Github, Paper-sWithCode, or academic repositories).

Task: The primary objective is to design, implement, and evaluate a custom solution to a complex Computer Vision problem. When asked to introduce a novelty, students often just stack more layers or add parameters to a network to artificially boost accuracy. To avoid this, your proposed novelty must be methodological and rigorously justified. **Simple image classification, multi-class classification, or basic regression tasks are strictly prohibited.** Acceptable domains include, but are not limited to: Semantic/Panoptic Segmentation, 3D Vision and NeRFs, Generative AI (GANs/Diffusion), Multimodal Vision-Language Models, Object Tracking, or Domain Adaptation. Students must base their work on at least **two peer-reviewed papers** (ideally from top-tier venues like CVPR, ICCV, ECCV, or NeurIPS), explicitly identify their limitations (e.g., computational inefficiency, poor generalization, boundary artifacts), and propose at least **one ad-hoc novelty** (e.g., a custom attention mechanism, a novel loss function, a unique data augmentation strategy, or an architectural optimization). The final submission must rigorously prove the effectiveness of this novelty.

Proposal Approval:

- *Mandatory Checkpoint:* Students choosing this project must submit a 1-page proposal summarizing their chosen topic, at least two reference papers, the identified limitations, and the proposed novelty via the Google Form by **May 10**. Professor approval is strictly required before writing any code to ensure the project scope is appropriate.

Main objectives:

- *Literature Review & Baseline:* Identify two relevant research papers, analyze their flaws/limitations, and successfully implement the baseline model(s) to reproduce their initial metrics.
- *Ad-Hoc Novelty Integration:* Formulate, implement, and integrate an original methodological novelty designed to overcome the identified limitations.
- *Ablation Study:* Conduct a rigorous ablation study to isolate the impact of your ad-hoc novelty. You must empirically demonstrate that your proposed change is the direct cause of any performance improvements over the baseline.
- *Strict Reproducibility:* Ensure the provided notebook/code repository contains fixed random seeds, clear environment requirements, and deterministic training scripts to guarantee 1-to-1 reproducibility of the results.

References:

- 1 To be provided by the student in their project presentation. (Minimum of 2 peer-reviewed papers mandatory).

Project 1: JPEG AI as a Threat to Deepfake Detection

Abstract: The emergence of JPEG AI, the first international standard for end-to-end learned image compression, introduces a new and underexplored challenge for multimedia forensics. Unlike traditional codecs, JPEG AI employs deep neural networks to encode and decode images, producing compression artifacts with a fundamentally different statistical signature from those of classical JPEG. Deepfake detectors are known to be sensitive to low-level frequency cues and compression artifacts embedded during the generation process; therefore, recompressing manipulated images with a neural codec may disrupt these cues and degrade detection performance. This project investigates the extent to which JPEG AI compression affects the reliability of state-of-the-art deepfake detectors, characterizes the nature of this degradation through frequency-domain analysis, and explores mitigation strategies to restore robustness under neural compression.

Dataset: Students can choose any image dataset

Task: The project is structured around three phases. In the first phase, students select a deepfake detection dataset and apply JPEG AI compression to a subset of the images at multiple quality levels (i.e., varying BPP), using the official [JPEG AI reference software](#). The compressed and uncompressed images are then used to benchmark different pre-trained deepfake detectors, measuring the degradation in detection performance as a function of compression strength. In the second phase, students conduct an in-depth analysis of why performance degrades: this involves inspecting frequency-domain representations of real, fake, and compressed images to understand how JPEG AI disrupts the artifact patterns that detectors rely upon. In the third phase, students investigate and compare mitigation strategies to recover detector robustness, such as fine-tuning on compression-augmented data, compression-aware preprocessing, or domain-adaptation techniques.

Main objectives:

- *Dataset preparation and compression pipeline:* Select a deepfake detection dataset and implement a reproducible pipeline that compresses images using JPEG AI at several target bitrates, preserving metadata for controlled experimentation.
- *Baseline evaluation under compression:* Evaluate one or more pre-trained deepfake detectors on both the original and JPEG AI-compressed test sets. Report standard metrics as a function of bitrate to characterize the degradation curve.
- *Frequency-domain forensic analysis:* Analyze how JPEG AI compression modifies the low-level statistical fingerprints that deepfake detectors exploit. Compare DCT coefficient distributions and power spectra between original, compressed-real, and compressed-fake images to explain the observed drop in detection accuracy.
- *Mitigation strategies:* Implement and compare at least two approaches to improve the robustness. Evaluate the trade-off between robustness gain and computational overhead.

References:

- E. D. Cannas et al., "Is JPEG AI Going to Change Image Forensics?," 2025 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), Honolulu, HI, USA, 2025, pp. 1575-1586, doi: 10.1109/ICCVW69036.2025.00167.

Project 2: Joint Detection of AI-Generated Images and Post-Processing Alterations in Real-World Scenarios

Abstract: AI-generated image detectors are typically evaluated under idealized conditions, where images are clean, uncompressed, and sourced directly from generative models. In practice, however, images circulating on social media have almost always undergone some form of post-processing each of which introduces a distinct degradation signature that can disrupt detector performance. This project proposes a unified multi-task deep learning framework that simultaneously addresses two forensic questions: whether an image is real or AI-generated, and which type of post-processing alteration has been applied to it. A further analytical objective investigates whether AI-generated and real images respond differently to the same post-processing operations, potentially revealing exploitable cross-class traces that can inform more robust detection strategies.

Dataset: **RRDataset:** a real-world robustness benchmark containing real photographs and AI-generated images with three splits: original, internet-transmitted, and re-digitized. Students may select a subset of the dataset, comprehensive of all three classes, according to their available hardware.

Task: Given a single input image, a unified deep learning model must produce two predictions simultaneously: (1) a binary real/fake label indicating whether the image is a genuine photograph or AI-generated, and (2) a multi-class transformation label indicating which post-processing alteration has been applied (original, internet-transmitted, or re-digitized). Both predictions are produced by a shared backbone, trained jointly with a combined loss function. The project is structured in three phases: baseline unimodal training, joint multi-task training, and an analytical phase investigating cross-class transformation traces.

Main objectives:

- *Data preparation and subset selection:* Download and preprocess the RRDataset, constructing a balanced split across real/fake classes and transformation categories. Students should document their hardware-driven subset choice and verify class balance before training.
- *Multi-task unified model:* Implement a shared backbone with two independent classification heads, one per task. Train the model jointly using a weighted combination of the two cross-entropy losses, and experiment with different loss weighting strategies. Evaluate both tasks simultaneously and compare against the unimodal baselines to assess whether joint training improves, degrades, or leaves unchanged the performance of each individual task.
- *Performance analysis across transformation types:* Break down real/fake detection accuracy separately for each transformation category, original, transmitted, and each re-digitization method, to identify which post-processing operations are most damaging to detection performance and whether the pattern differs between real and AI-generated images.
- *Ablation study:* Analyze how different weightings of the two task losses affect the trade-off between real/fake accuracy and transformation classification accuracy, identifying whether the two tasks compete for representational capacity or complement each other.

References:

- Li, Chunxiao, et al. "Bridging the Gap Between Ideal and Real-world Evaluation: Benchmarking AI-Generated Image Detection in Challenging Scenarios." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2025.
- Shao, Rui, Tianxing Wu, and Ziwei Liu. "Detecting and grounding multi-modal media manipulation." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2023.

Project 3: Topologically-Constrained Multi-Task Learning for Periodontitis Grading

Abstract: Deep learning models for Periodontal Bone Loss (PBL) often struggle with generalizability because they treat classification as a global image task, occasionally ignoring the rigid anatomical relationship between the Cementoenamel Junction (CEJ) and the Alveolar Bone Crest (ABC/AC). While multi-task models exist, they often lack a geometric link between segmentation and classification branches, potentially leading to anatomically inconsistent predictions. This project explores the design of a network where the segmentation of those landmarks explicitly constrains the classification logic, either through direct geometric rules or via learned latent representations that map the clinical anatomy to a constrained feature space. This ensures that the model adheres to clinical landmarks rather than image artifacts.

Dataset: DenPAR: Intra-Oral Periapical Radiographs Dataset.

Task: This project aims to develop a multi-task network, by using architectures such as Mask R-CNN or a dual-head HRNet, to perform simultaneous semantic segmentation of dental structures and classification of periodontitis stages. The methodological focus is on implementing an ad-hoc novelty that enforces anatomical consistency, which could be for example a *Distance-Map Loss* to penalize topological inconsistencies or a *Graph Convolutional Module* to model the CEJ-ABC relationship from a graph perspective. Essentially, by forcing the model to perform classification alongside segmentation, the model gets forced to learn these important clinical landmarks. The final system will be evaluated on its ability to produce accurate clinical indicators via segmentation while maintaining high classification precision.

Main objectives:

- *Baseline Implementation:* Implement a standard multi-task independent dual-branch architecture for dental segmentation and classification to establish a baseline for performance and error analysis.
- *Geometric Constraint Integration:* Develop and integrate an original methodological improvement, such as a landmark-aware attention module, a custom loss function, a graph-based constraint, or a latent representation of the relationship to align classification with anatomical distances.
- *Interpretability Analysis:* Utilize tools like Grad-CAM to verify that the model's focus has transitioned from background noise or irrelevant features to relevant clinical indicators derived from the segmentation branch.
- *Ablation Study:* Compare the classification accuracy and anatomical feasibility of the baseline model against the constrained version.

References:

- 1 Do KQ, et al. "Development and validation of artificial intelligence models for automated periodontitis staging and grading using panoramic radiographs." BMC Oral Health. 2025 Oct 14; doi: 10.1186/s12903-025-07025-8.
- 2 Oktay, O., et al. "Anatomically Constrained Neural Networks (ACNNs): Application to Cardiac Image Enhancement and Segmentation." IEEE TMI, 2018.
- 3 Kendall, A., et al. "Multi-Task Learning Using Uncertainty to Weigh Losses for Scene Geometry and Semantics." CVPR, 2018.

Project 4: Anatomically-Constrained LDMs for Rare Case Augmentation

Abstract: Deep learning models for Periodontitis often suffer from severe class imbalance, as "Stage IV" (severe) cases are significantly rarer than "Stage I" cases in public datasets. While Generative Models can synthesize data, standard Denoising Diffusion Probabilistic Models (DDPMs) often suffer from anatomical hallucinations, where the generated images look realistic but lack clinically correct tooth and bone structures. This project investigates how to use Latent Diffusion Models to generate rare clinical cases that are strictly constrained by anatomical masks to ensure clinical validity.

Dataset: **Periodontal Bone Loss Keypoint and Detection Dataset** (includes Bounding Boxes, Keypoints, and Instance Segmentation masks).

Task: The project involves developing a 2D-LDM pipeline to synthesize high-quality dental X-ray patches for severely underrepresented classes. Students will implement methodological enhancements to ensure the generated textures align with real-world anatomical structures. The work includes designing a guidance mechanism, such as a mask-conditioning branch using segmentation masks or an anatomical-based loss, to keep bone levels and lesions geometrically consistent with the tooth root. The effectiveness of the synthetic data will be validated through standard generative models evaluation metrics and by measuring its impact on a downstream classification and/or detection model.

Main objectives:

- *Baseline LDM Implementation:* Implement a standard 2D Diffusion model (e.g. LDM) trained on localized, high-resolution cropped patches of rare lesions, to generate and identify common structural errors in unconstrained generation.
- *Anatomical Guidance Integration:* Formulate and integrate an original guidance mechanism to maintain strict clinical consistency between the generated bone levels and actual anatomical structures.
- *Downstream Impact Evaluation:* Artificially augment the training set by importing the synthesized, anatomically-correct minority class in the imbalanced dataset. Compare the Mean Average Precision (mAP) of a downstream model trained on real data against the (real+)augmented data, evaluating on a strict holdout test set.
- *Ablation Studies:* Conduct an ablation study to analyze the trade-off between the strength of the anatomical mask constraints and the diversity of the generated imagery, as well as a possible comparison with a DDPM.

References:

- 1 Ho, J., Jain, A., & Abbeel, P. "Denoising Diffusion Probabilistic Models". NeurIPS 2020.
- 2 Kazerooni, A., et al. "Diffusion models in medical imaging: A comprehensive survey", Medical Image Analysis 2023. DOI: 10.1016/j.media.2023.102846
- 3 Pinaya, W. H. L., et al. "Brain Imaging Generation with Latent Diffusion Models". MICCAI 2022.
- 4 Konz, N., et al. "Anatomically-Controllable Medical Image Generation with Segmentation-Guided Diffusion Models". MICCAI 2024.

Project 5: Lightweight Diffusion Models: Accelerating Training/Inference for Resource-Constrained Environments

Abstract: Denoising Diffusion Probabilistic Models (DDPMs) achieve state-of-the-art results in image generation, but their iterative sampling process is notoriously slow and computationally expensive. This makes them difficult to deploy on edge devices or train in resource-constrained environments. This project explores methodologies to drastically reduce the computational burden and inference time of diffusion models without severely degrading the generation quality.

Dataset: CIFAR-10, Oxford 102 Flowers, CelebA. (*Or what you can run on your hardware*)

Task: The objective is to implement a baseline DDPM and subsequently integrate a methodological novelty designed to accelerate the sampling process or reduce the model's footprint. Students cannot simply reduce the number of parameters by blindly cutting layers. Acceptable approaches include implementing Denoising Diffusion Implicit Models (DDIM) for deterministic fast sampling, latent space diffusion (training an autoencoder first to diffuse in a smaller latent space), or knowledge distillation techniques. The project must rigorously evaluate the trade-off between computational efficiency (e.g., sampling time) and generation quality (FID metric).

Main objectives:

- *Baseline DDPM Implementation:* Train a standard DDPM on a low-resolution dataset (or what your hardware can run) and measure the baseline Fréchet Inception Distance (FID) or Inception Score (IS), alongside the average inference time for generating a batch of images.
- *Anatomical Guidance Integration:* Formulate and integrate an algorithmic improvement (e.g., DDIM sampling strategy, a latent-space approach, progressive distillation, or something new!) aimed at accelerating inference or training.
- *Downstream Impact Evaluation:* Quantify the speed-up factor and compare the visual quality and quantitative metrics of the accelerated model against the baseline.
- *Fidelity vs. Diversity Study:* Conduct an ablation study on the number of sampling steps (e.g., $T = 1000$ vs. $T = 100$ vs. $T = 10$). Analyze how the proposed efficiency method handles severe step reductions compared to the standard DDPM scheduler.

References:

- 1 Song, J., Meng, C., & Ermon, S. "Denoising Diffusion Implicit Models." ICLR 2021.
- 2 Rombach, R., et al. "High-Resolution Image Synthesis with Latent Diffusion Models." CVPR 2022.
- 3 Salimans, T., & Ho, J. "Progressive Distillation for Fast Sampling of Diffusion Models." ICLR 2022.

Project 6: Emerging Interpretability in LeJEPA vs. Supervised Learning

Abstract: This project investigates the emergence of "zero-shot" semantic segmentation in Joint-Embedding Predictive Architectures (JEPAs) compared to traditional supervised models. While supervised training is driven by categorical labels, LeJEPA utilizes Sketched Isotropic Gaussian Regularization (SIGReg) to learn representations. We aim to evaluate whether the semantic structures emerging in LeJEPA's latent space (visible via PCA) align more naturally with the model's internal attention and gradients than those in supervised counterparts.

Dataset: ImageNet-1K, ImageNet-100, CIFAR-10, CIFAR-100.

Task: A comparative layer-by-layer analysis of feature localization and semantic consistency between LeJEPA-trained models and End-to-End Supervised models, utilizing both Vision Transformers (ViT) and Convolutional Neural Networks (CNNs).

Main objectives:

- *Model Training and Feature Extraction:* Deploy and contrast ViT and CNN architectures trained via LeJEPA and standard supervised learning. Extract latent representations across different layers for comparative probing.
- *Mapping XAI and Semantic Projections:* Implement a retrieval pipeline for explainable ai (XAI) saliency maps (using GMAR or Attention Rollout for ViTs and Grad-CAM for CNNs) and compute semantic masks through PCA projections of the embedding space.
- *Definition of a Semantic Alignment Score (SAS):* Design and implement a quantitative framework to measure the association between XAI-based saliency and PCA-based semantic components. The student should explore and justify the choice of metrics (e.g., spatial correlation, mutual information, or localized overlap) to define a robust score.
- *Qualitative and Comparative Analysis:* Conduct an in-depth visual study to assess if LeJEPA's "free" semantic segmentation provides a more interpretable representation of objects compared to the potentially biased focus of supervised models.

References:

- Balestrieri, R., & LeCun, Y. (2025). *LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics*.
- Jo, S., Jang, G., & Park, H. (2025). *GMAR: Gradient-Driven Multi-Head Attention Rollout for Vision Transformer Interpretability*.
- Selvaraju, R. R., et al. (2017). *Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*.

Project 7: Lean Geometry - Boosting Mobile MDE with LeJEPA

Abstract: High-quality Monocular Depth Estimation (MDE) typically relies on heavy backbones and massive supervision. This project explores whether the LeJEPA self-supervised objective can enhance the geometric representation of lightweight models like METER (Mobile Vision Transformer for MDE). By substituting standard supervised pre-training with LeJEPA's Sketched Isotropic Gaussian Regularization (SIGReg), we investigate if a mobile-grade model can "inherit" the dense feature quality of larger world models while remaining suitable for real-time inference on edge devices.

Dataset: NYU Depth V2, KITTI.

Task: Comparative study of a lightweight MDE architecture (METER/MobileViT) pre-trained with LeJEPA vs. ImageNet-Supervised pre-training, focusing on depth accuracy and feature interpretability.

Main objectives:

- *Lightweight LeJEPA Integration:* Implement the LeJEPA training objective (SIGReg) on a mobile-friendly backbone (e.g., the ViT-lite used in METER). The goal is to perform self-supervised pre-training on static images with minimal computational resources (single GPU).
- *Zero-shot Geometric Probing:* Before adding the depth decoder, analyze the latent space using PCA. Evaluate if the "free" semantic segments emerging from LeJEPA align with depth discontinuities (e.g., if the PCA map naturally separates foreground objects from the background).
- *MDE Downstream Performance:* Fine-tune the LeJEPA-METER model for depth estimation. Compare the results against the original supervised METER in terms of RMSE, AbsRel, and δ_1 metrics, verifying if SSL pre-training speeds up convergence.
- *Efficiency-Accuracy Trade-off:* Quantify the performance gains relative to the training cost. Analyze if LeJEPA's "heuristics-free" approach (no teacher-student, no stop-gradient) makes it more suitable for students to adapt MDE models to specific robotic domains with limited data.

References:

- Balestrieri, R., & LeCun, Y. (2025). *LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics*.
- Papa, L., Russo, P., & Amerini, I. (2024). *METER: A Mobile Vision Transformer Architecture for Monocular Depth Estimation*.
- Mur-Labadia, L., et al. (2026). *V-JEPA 2.1: Unlocking Dense Features in Video Self-Supervised Learning*. (Per il contesto sulle dense features).